Future City Fellowship — Milestone 2 Final Report

Safe, Code-Responsive Electrification for Fairfax City and Prince William County

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Supersedes: `Future_City_Fellowship_Milestone_2_Phase_Report.md` (July 2026) — that report remains the authoritative record of the Phase 2 build; this report integrates the gap-closure implementation completed 2026-07-19.

> ***Standing discipline.*** *This project is a planning-support prototype. Nothing in this report is an engineering design, permit or code determination, utility capacity confirmation, official forecast, budget, procurement package, or construction authorization.* ***0 of 12 candidate sites have High data confidence; electrical capacity is a placeholder at all 12 sites.*** *No client endorsement, AHJ acceptance, utility confirmation, or funding award exists or is implied.*

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1. Executive summary

Phase 1 identified a local-readiness problem: electrification is expanding faster than code interpretation, responder training, siting practice, and emergency-access planning in two contrasting Dominion-served jurisdictions — built-out Fairfax City (~25k) and county-scale Prince William County (~500k, dominated by data-center load growth). Phase 2 answered it with a reproducible decision-support system; the July 2026 compliance audit then identified exactly what Milestone 2 still required. This report documents the **implementation run of 2026-07-19** that closed every gap that code, modeling, analysis, documentation, or artifact production can close — and prepared complete, ready-to-execute packages for every gap that only the client, team, AHJ, utility, facilities staff, or fellowship staff can close.

The leading solution (research-ranked, pending team adoption): distributed staged pilots at the strongest candidate sites, with expandable civil/electrical pathways, monitoring of actual use, and activation of additional chargers only when predefined, equity-aware triggers are sustained. It scored 85/100 against six genuine alternatives, wins on both evidence streams (the client's own overbuild/trenching cost history and the model's low simulated utilization), and grows naturally into the hybrid portfolio (81/100) as ports multiply. The county's ~35-station fleet hub remains a coordinated parallel track, not a competitor.

What is new since the Phase 2 report (all verified this run):

- an **independently reproduced baseline** — the full 25-step V2 workflow re-run with every check passing (FFX002 top at 58.8; 25 starter / 48 pathway ports; 38 REVIEW + 12 INFO findings; 15 Tableau exports; 35-file package);

- a **V3 model** (`fcf_m2` 3.0.0) with two permanent modes: `legacy_v2` (bit-exact baseline reproduction, golden-tested) and `milestone2_v3` (canonical 2026 + 2036/2046/2056 horizons, fixed-anchor cohort-independent scoring with a migration report, typed data confidence, peak-window and discrete-event wait modeling, lifecycle economics, emissions, proposed equity-aware expansion triggers); 80 tests, 92% coverage;
- one **canonical cross-model data contract** (hybrid CSV + JSON, v1.0.0, 120 scenario rows, schema-validated and hash-chained) now consumed by the Excel workbook, the MATLAB adapter, the Tableau patch package, the static dashboard, and this report — ending the "three truths" problem;
- **two aspects modeled at 2036/2046/2056** (§5);
- a **lifecycle-economics layer**: staged vs broad-buildout vs defer, net present cost with avoided-cost headline (ROI deliberately rejected — no honest revenue line exists);
- a **net-CO2e layer** with a corrected, sourced grid factor (EPA eGRID2023 Rev 2, SRVC, **596.3 lb CO2e/MWh**, accessed 2026-07-19 — the research package's 590.2 figure was superseded by direct EPA verification) plus the avoided-EVSE resource-management metric;
- a **standards package** (UL 2594 + UL 3600 highlighted; UL 2231/NEC 625-as-adopted/VA USBC-SFPC/Access Board supporting; conditional standards explicitly out of scope) and a **proposed standard concept** with seven test modules and TBV-disciplined parameters;
- an updated **Excel workbook** (pie chart repaired, horizons to 2056, contract import, cost/CO2e sheets, live reconciliation checks — all PASS);
- a rebuilt **MATLAB/Simulink repository** implementing all twelve improvement items as source, tests, and documentation — with execution honestly **blocked** (no licensed MATLAB available) and exact commands prepared for a licensed runner;
- a complete **presentation package**, **static offline dashboard**, **website/assistant patch package**, and **six human-action packages** (client meeting #2, team session, AHJ, utility/facility data, public engagement, fellowship clarification).

Honest status of the two aspects at a glance (screening values, base escalation, contract v1.0.0):

Horizon	Active ports (base→high)	Staged NPC	Broad NPC	Staged avoids	Net CO2e avoided (base→high)
2026	25	—	—	—	≈43 t/yr
2036	44 → 47	$1.56M	$1.87M	**$0.31M**	≈82 → 113 t/yr
2046	47 → 48	$2.29M	$2.64M	**$0.34M**	≈130 → 195 t/yr
2056	47 → 48 (pathway ceiling 48)	$3.01M	$3.41M	**$0.39M**	≈173 → 260 t/yr

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2. Verified baseline (what this report stands on)

Every headline number was independently re-derived on 2026-07-19 by re-running the workflows on disposable copies:

Check	Result
Python V1 (`run_all.py` + checker)	PASSED
Python V2 — 25 steps, output checker, baseline comparison, Tableau exports, package	ALL PASS: 12 sites; FFX002 City of Fairfax Regional Library #1 at 58.8 (Medium); FFX004 54.4; PWC006 52.1; 25 starter / 35 / 44 / 44 / 54 ports across legacy cases; 48 future-

	ready; 38 REVIEW + 12 INFO; 15/15 exports; 35/35 package files
High-confidence sites	**0 of 12** (preserved as the project's credibility signature)
Offline GIS	Cold-start offline fails (documented V2 defect); V3 fix: cached VA TIGER boundaries in-repo — map generation verified with networking disabled
MATLAB	Static inspection only. **No simulation executed — none claimed.**

The V3 `legacy_v2` mode reproduces the accepted baseline **bit-for-bit** (golden tests assert equality with the frozen V2 outputs on every score, rank, port count, and strategy string).

3. Client evidence and the honest Client Meeting #2 status

The [client contact] interview supplies the project's strongest operational evidence: McCoart's 10-port install cost escalated through trenching/undergrounding distance; the Neabsco garage's 70 ports were heavily underused and 17 stations were taken offline at a ~$1,000/station/yr operating frame (~$20k/yr avoided); Virginia's "sufficient EV charging" requirement is ambiguous (LEED ~5% used as guidance); the Animal Services Center satisfied the EV-ready requirement with conduit only; ADA EV-space requirements forced a year of fleet-hub redesign (Dominion reports identical friction). [fire-service interviewee] (PWC Fire & Rescue/HazMat) established the proactive window: **no local EV fires yet**, but recognition/education and charger siting that can block responder access are live concerns.

Client Meeting #2 has not yet occurred as defined by the program. The interview was an information-gathering session; the program's CM2 requires presenting the narrowed focus, brainstorming solutions with the client, and capturing feedback on the team's actual alternatives. explicitly offered a follow-up ("either one, follow up meeting, or emails is fine"). The complete follow-up package — ready-to-send email draft, two-page pre-read, 30-minute agenda, alternatives briefing, feedback form, decision log, follow-up email, and acceptance checklist — is prepared in `human_actions/client_meeting_2/`. **Nothing in this report converts that preparation into a held meeting or a client endorsement.**

4. Alternatives and convergence (prepared for team adoption)

Seven genuinely distinct alternatives were developed and scored in a provisional MCDA (weights 25/20/20/15/10/10 across safety-code readiness, demand fit, lifecycle cost/funding, expandability/resilience, equity, data confidence):

Alternative	Score
B — Distributed staged pilots + expandable pathways	**85**
F — Hybrid portfolio (B + EV-ready seeding + managed charging)	81
E — Managed-charging-first	72
C — EV-ready-first (pathways only)	71
D — Centralized fleet hub + selective public (county's real parallel track)	58
A — Broad immediate buildout (the Neabsco pattern)	41
G — Defer / do nothing (comparator)	37

Sensitivity: the ranking survives ±5pp on every criterion; the one honest crossover is equity-dominant weighting (~2× default), where F overtakes B — neutralized by presenting staged pilots as the hybrid's first stage. **These scores are researched provisional values; the team session package (`human_actions/team_alternatives_session/`) exists precisely so the team can amend and adopt them honestly. No team adoption is claimed.**

5. Two aspects at 2036 / 2046 / 2056

The official requirement — two aspects modeled at 10/20/30-year timelines — is satisfied with the canonical convention 2036/2046/2056 from a stated 2026 baseline (exact calendar years are a convention, not a mandate; the fellowship-clarification draft asks for confirmation).

Aspect 1 — site/portfolio charging demand and port/pathway capacity. The Python model of record, extended by growth cases (base/high/stress at each horizon; multipliers 1.9/2.6 → 3.0/4.5 → 4.0/6.0, stress ×1.15 — provisional conventions pending calibration) with saturation capped by the 48-port pathway and monotonic activation. Peak-window Erlang and a seeded 400-day discrete-event simulation add P10/P50/P90 wait bands; at current demand even peak-window P90 waits sit under the 10-minute target at every top site.

Aspect 2 — lifecycle cost and avoided overbuild, carrying the environmental outcome. Full cash-flow model 2026–2056 per site (hardware, civil with contingency, pathway premium, trigger-year expansions, network fees on active ports only, maintenance, electricity, ~year-10 electronics replacement, decommissioning of never-needed ports in the overbuild case; low/base/high escalation; 3% real discount plus undiscounted view). Result: **the staged plan costs less than broad buildout at every horizon under every escalation case**, with the avoided-cost headline growing from $0.31M (2036) to $0.39M (2056) on the base case. The attached environmental outcome: net avoided CO2e (displaced tailpipe minus charging grid emissions, eGRID2023r2 SRVC 596.3 lb CO2e/MWh with vintage stated everywhere) reaching ≈173–260 t/yr by 2056, paired with the avoided-EVSE metric (13–23 pathway ports deliberately not energized on the base path — the direct answer to the client's decommissioning experience).

Success evidence per horizon is defined (2036: sustained trigger-level utilization at ≥2 pilots, documented first expansion decision; 2046: portfolio utilization within band, cost/port vs projection; 2056: adopted-share demand served without exceeding pathway capacity).

6. Canonical contract and cross-model integration

`data_contract/v1/` (schema.json, scenario_table.csv — 120 rows, site_table.csv, source_provenance.csv, standards_register.csv, assumptions.json, contract_manifest.json) is the single source of truth. Python generates it; Excel, MATLAB, Tableau exports, the static dashboard, and the presentation consume it. Validation implements the ten reconciliation tests (identity joins on `site_id` only, completeness, monotonicity, legacy totals 25/48, unit ranges, year-set equality, placeholder propagation, energy accounting, factor-vintage pinning, manifest hash chain) — all passing; deliberately corrupted contracts fail every corresponding test. The single most important field is **`capacity_basis`**: it makes placeholder-vs-verified machine-readable, so no downstream artifact can silently present placeholder capacity as real. The MATLAB adapter is now generated (PY-14), eliminating the hand-built CSV and its 2050-base/high port mismatch.

7. Model integrity upgrades (V3)

- **Cohort-independent scoring (PY-02):** versioned fixed anchors; the same site scores identically when unrelated sites are added or removed (tested). Ranking differences vs the cohort-relative baseline are published in a migration report; the port/pathway structure remains anchored to the accepted legacy baseline pending deliberate re-baselining.
- **Typed data confidence (PY-05):** field-level provenance with controlled source classes; note wording can no longer move confidence scores (tested). Overall confidence is traceable to a per-field breakdown. Result unchanged in substance: 0 High-confidence sites.
- **Demand honesty (PY-03):** `demand_model_status = synthetic_uncalibrated` is embedded in the contract; the calibration routine imports observed sessions, fits with a holdout split, and **refuses** the calibrated label on insufficient data or poor fit (tested).
- **Equity-aware expansion triggers (PY-20):** proposed policy — sustained utilization ≥40% or repeated 10-minute waits, ≥97% uptime before data are trusted, ≥6 months observation, pathway headroom, budget gate requiring human approval, and an **equity override** lowering the utilization gate to 25% in high-equity-need areas so low-adoption communities are not permanently starved. Every evaluation emits a full rule trace; current status everywhere: `awaiting_observed_data`. **The policy is proposed configuration, not adopted municipal policy.**

8. Standards and the proposed standard

Highlighted: UL 2594 (EVSE safety, 3rd ed. 2022) and UL 3600 (circular-economy measurement, 2023) — the only pair satisfying both the safety and sustainability rubric rows. **Essential supporting:** UL 2231-1/-2; NEC Article 625 **as adopted in Virginia (2020 NEC via the 2021 USBC, effective 2024-01-18** — the 2023 NEC is not Virginia law); Virginia USBC/SFPC (2024 code cycle underway, ~2027); Access Board EVSE guidance (**still at NPRM as of 2026-07-19**; VA USBC ADA provisions govern today); OCPP/ISO 15118 as supporting protocols (not ULSE standards). **Conditional and out of scope:** UL 2202 (no DCFC), UL 9540/9540A + NFPA 855 (no storage; `battery_storage_flag=0` at all sites — and therefore **no resilience claim is made anywhere**).

The proposed standard — *Municipal EV Charging Site Readiness and Emergency Access Standard*, explicitly a **fellowship design concept, not an official UL/ULSE proposal** — defines seven test modules (emergency access/disconnect; expandability/right-sizing documentation; accessibility cross-references; monitoring/triggers; managed-charging fail-safe; commissioning evidence; materials/end-of-life) with three conformance levels. Sourced parameters cite adopted codes by edition; every unsourced parameter is marked **TBV by ULSE/AHJ/manufacturer/adopted code**. Full texts: `reports/FINAL_STANDARDS_APPLICATION_MATRIX.md/.docx` and `reports/PROPOSED_MUNICIPAL_EV_STANDARD_CONCEPT.md/.docx`.

9. Technical integration, risk, environment, funding, stakeholders

- **Integration:** the full chain (utility → transformer/feeder → service entrance/panel → load management → EVSE → vehicles, with OCPP/ISO 15118 communications, metering, emergency disconnect and responder access, monitoring feeding the governance/trigger layer, and the county fleet hub as coordination-only) is documented with an 8-interface register and responsibility register (`reports/TECHNICAL_INTEGRATION_ARCHITECTURE.md`, diagram in `visuals/`).

- **Risk:** an 18-risk register with likelihood×consequence, owner, trigger, mitigation, evidence, residual risk, response time, and deciding human authority (`reports/RISK_REGISTER.md/.csv`). Top mitigations: fail-safe curtailment on comms loss; verified/labeled/illuminated disconnect at commissioning; service-capacity confirmation before design freeze; dual-path utilization logging; ADA-first layout.
- **Environment/resilience:** heat, flood, stormwater, outage, and severe-weather treatment; sourced grid factor with annual refresh; materials/take-back/UL 3600 procurement line; trenching minimization doubling as embodied-impact minimization; explicit no-resilience-claim statement (`reports/ENVIRONMENT_AND_RESILIENCE.md`).
- **Funding/budget:** FY2027 context (Fairfax City ≈$207.5M GF adopted 2026-05-05; PWC ≈$1.955B GF revenues adopted 2026-04-21 — a 2-port pilot is CIP-scale-trivial but must ride a process); Dominion Level 2 Charging Program (50% make-ready, 100% for qualifying disadvantaged communities) as the recommended first path; NEVI presented only with its volatility history ($885M FY26; VA solicitations anticipated Q1 2026); CFI watchlist. **No funding is implied as awarded** (`reports/ECONOMIC_AND_FUNDING_BRIEF.md`).
- **Stakeholders/regulatory:** the nine-lane jurisdictional map naming the deciding offices per Report 10's verified table (`reports/REGULATORY_STAKEHOLDER_MAP.md`); public-engagement instruments (survey, accessibility review, listening sessions, language access, VCDPA-aware privacy, feedback-to-decision traceability) in `human_actions/public_engagement/`.

10. Artifact status

Artifact	Status
Excel workbook	`excel/Electric_Data_Milestone_2_Updated.xlsx`: pie chart repaired and connected; proxies labeled; extended to 2056 (labeled extrapolation); canonical import; lifecycle + CO2e sheets; low/base/high; assumptions register; **six live reconciliation checks all PASS**; renders without repair warnings. Original preserved.
MATLAB/Simulink	`repos/matlab_simulink_m2_v3/`: all 12 items implemented as source/tests/docs; static checks pass; **execution BLOCKED (no license)**; exact licensed-runner commands provided. Legacy repo preserved; committed legacy visuals usable only with the placeholder-capacity caption.
Tableau	Verified `.twbx` **blocked** (no Tableau Desktop; Tableau Public rejected because it publishes). Full blocker note, contract-derived patch CSVs, and an exact build recipe in `tableau/`; the presentation-facing deliverable is the static dashboard (the team's own chosen demo strategy).
Static visuals	9 contract-derived, footer-labeled, offline-openable visuals + combined PDF (`visuals/static_dashboard/`); automated footer QA passes.
Website/assistant	Offline patch package with file-by-file guide, privacy notice, authority disclaimers, source-vintage footer, assistant boundaries, deployment/rollback runbook (`website_patch_package/`). **Nothing deployed.**
Presentation	Final deck, PDF, speaker notes, ≤13-minute timing runbook, Q&A backup, self-contained static visuals (`presentation/`), rendered and inspected slide-by-slide.

11. Limitations (unchanged in force)

1. 0 of 12 sites have High data confidence; every electrical field is unconfirmed.
2. Capacity is a placeholder at 12/12 sites; no feasibility claim is permitted on it.
3. Demand is synthetic/uncalibrated; growth multipliers are provisional conventions.
4. All costs are screening-class; the $429,500-class screens are not budgets.
5. Emissions rest on labeled benchmark assumptions and an annually-refreshed factor.
6. Client Meeting #2 is prepared, not held; no client endorsement exists.
7. Team adoption of alternatives/MCDA/conventions is prepared, not performed.
8. MATLAB execution and Tableau workbook verification are blocked by tooling, stated plainly.
9. No AHJ, utility, facility, or fellowship confirmation of anything has occurred.

12. Next human actions (highest priority first)

10. **Send the CM2 follow-up request to** [client contact] and hold the 30-minute touchpoint or email exchange before July 21 (`human_actions/client_meeting_2/01_EMAIL_DRAFT_TO_CLIENT.md`).
11. **Hold the team alternatives session** (60–90 min) and adopt/amend the MCDA honestly; assign Paula's presentation role (`human_actions/team_alternatives_session/`).
12. **Send the fellowship clarification email** (CM2 classification; year convention; standards count) (`human_actions/fellowship_confirmation/`).
13. **Send the utility/facility data requests** for FFX002/FFX004/PWC006 (`human_actions/utility_facility_data/`) — the single highest-value unlock (upgrades `capacity_basis` from placeholder).
14. **Schedule Fire Marshal/AHJ conversations** in both jurisdictions with the prepared question sets (`human_actions/ahj_fire_marshal/`).
15. Rehearse twice to ≤13 minutes (Monday evening + Tuesday morning per the timing runbook).
16. Post-M2: run the MATLAB batch on a licensed installation; build/verify the Tableau workbook per the patch recipe; deploy the website patch per the runbook; field the engagement instruments; begin Dominion program application alignment.

13. Peer-team question

> ***How should a city set an equitable expansion trigger for public EV charging when early utilization data are sparse — so it avoids both overbuilding and under-serving communities whose current EV adoption may be lower?***

Backups: (1) what evidence should convert EV-ready conduit into active ports, and who owns that decision; (2) how to keep a 30-year model honest when 10-year goals are already off-track (phrase with care).

Implementation integrity: no original source file, repository, or external system was modified; nothing was emailed, pushed, deployed, or published; no client, team, AHJ, utility, facility, fellowship, or MATLAB-execution evidence was fabricated. Full manifests: `00_IMPLEMENTATION_MANIFEST.md`, `verification/`.